In [1]:
```
import pandas as pd
```

In [174…
```
df = pd.read_csv(r'C:\Users\pandi\Desktop\Abhiraj\Data Science\Notes\Sql Dataset\da
```

In [50]:
```
df
```

Out[50]:

	destination	passanger	weather	temperature	time	coupon	expiration	g
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	F
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	F
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	F
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	F
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	F
...	...	...	...	...	...	...	...	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	
12682	Work	Alone	Snowy	30	7AM	Bar	1d	
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	

12684 rows × 27 columns

In [25]:
```
df.columns
```

Out[25]:
```
Index(['destination', 'passanger', 'weather', 'temperature', 'time', 'coupon',
       'expiration', 'gender', 'age', 'maritalStatus', 'has_children',
       'education', 'occupation', 'income', 'car', 'Bar', 'CoffeeHouse',
       'CarryAway', 'RestaurantLessThan20', 'Restaurant20To50',
       'toCoupon_GEQ5min', 'toCoupon_GEQ15min', 'toCoupon_GEQ25min',
       'direction_same', 'direction_opp', 'Y', 'row_count'],
      dtype='object')
```

In [26]:
```
df[['weather','temperature']]
```

Out[26]:

	weather	temperature
0	Sunny	55
1	Sunny	80
2	Sunny	80
3	Sunny	80
4	Sunny	80
...	...	...
12679	Rainy	55
12680	Rainy	55
12681	Snowy	30
12682	Snowy	30
12683	Sunny	80

12684 rows × 2 columns

In [12]:

```
df.head(10)
```

Out[12]:

	destination	passanger	weather	temperature	time	coupon	expiration	gende
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Femal
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Femal
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Femal
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Femal
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Femal
5	No Urgent Place	Friend(s)	Sunny	80	6PM	Restaurant(<20)	2h	Femal
6	No Urgent Place	Friend(s)	Sunny	55	2PM	Carry out & Take away	1d	Femal
7	No Urgent Place	Kid(s)	Sunny	80	10AM	Restaurant(<20)	2h	Femal
8	No Urgent Place	Kid(s)	Sunny	80	10AM	Carry out & Take away	2h	Femal
9	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	1d	Femal

10 rows × 27 columns

In [27]:

```
df['passanger'].unique()
```

Out[27]:

```
array(['Alone', 'Friend(s)', 'Kid(s)', 'Partner'], dtype=object)
```

In [33]:

```
df[df['destination']=='Home']
```

Out[33]:

	destination	passanger	weather	temperature	time	coupon	expiration	g
13	Home	Alone	Sunny	55	6PM	Bar	1d	F
14	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	F
15	Home	Alone	Sunny	80	6PM	Coffee House	2h	F
35	Home	Alone	Sunny	55	6PM	Bar	1d	
36	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	
...	...	...	...	...	...	...	...	
12675	Home	Alone	Snowy	30	10PM	Coffee House	2h	
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)	1d	
12677	Home	Partner	Sunny	30	6PM	Restaurant(<20)	1d	
12678	Home	Partner	Sunny	30	10PM	Restaurant(<20)	2h	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	

3237 rows × 27 columns

In [37]:

```
df.sort_values('coupon')
```

Out[37]:

	destination	passanger	weather	temperature	time	coupon	expiration	g
11702	Home	Partner	Sunny	30	10PM	Bar	2h	F
9930	No Urgent Place	Alone	Snowy	30	2PM	Bar	1d	F
10632	Home	Alone	Rainy	55	6PM	Bar	1d	
7997	No Urgent Place	Friend(s)	Rainy	55	10PM	Bar	2h	
11166	Work	Alone	Snowy	30	7AM	Bar	1d	F
...	...	...	...	...	...	...	...	
10476	Home	Alone	Sunny	80	6PM	Restaurant(<20)	1d	F
5447	Home	Alone	Sunny	80	10PM	Restaurant(<20)	2h	F
10478	Home	Alone	Snowy	30	10PM	Restaurant(<20)	2h	F
5440	No Urgent Place	Alone	Sunny	80	2PM	Restaurant(<20)	2h	F
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	F

12684 rows × 27 columns

In [55]:

```
df.rename(columns={'destination':'dest'},inplace = True)
#if you want to modify the column paramently apply inplace = True
```

In [56]:

```
df
```

Out[56]:

	dest	passanger	weather	temperature	time	coupon	expiration	gende
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Femal
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Femal
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Femal
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Femal
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Femal
...	...	...	...	...	...	...	...	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	Mal
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	Mal
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	Mal
12682	Work	Alone	Snowy	30	7AM	Bar	1d	Mal
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	Mal

12684 rows × 27 columns

In [57]:

```
df
```

Out[57]:

	dest	passanger	weather	temperature	time	coupon	expiration	gende
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Femal
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Femal
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Femal
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Femal
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Femal
...	...	...	...	...	...	...	...	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	Mal
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	Mal
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	Mal
12682	Work	Alone	Snowy	30	7AM	Bar	1d	Mal
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	Mal

12684 rows × 27 columns

In [77]:

```
df.groupby(['occupation']).size() # this prove the data with count
df.groupby(['occupation']).size().to_frame('count') # to_frame-> added the count co
df.groupby(['occupation']).size().to_frame('count').reset_index() # reset_index -->
```

Out[77]:

	occupation	count
0	Architecture & Engineering	175
1	Arts Design Entertainment Sports & Media	629
2	Building & Grounds Cleaning & Maintenance	44
3	Business & Financial	544
4	Community & Social Services	241
5	Computer & Mathematical	1408
6	Construction & Extraction	154
7	Education&Training&Library	943
8	Farming Fishing & Forestry	43
9	Food Preparation & Serving Related	298
10	Healthcare Practitioners & Technical	244
11	Healthcare Support	242
12	Installation Maintenance & Repair	133
13	Legal	219
14	Life Physical Social Science	170
15	Management	838
16	Office & Administrative Support	639
17	Personal Care & Service	175
18	Production Occupations	110
19	Protective Service	175
20	Retired	495
21	Sales & Related	1093
22	Student	1584
23	Transportation & Material Moving	218
24	Unemployed	1870

In [91]:
```
df.groupby('weather')['temperature'].mean().to_frame('avg_temp').reset_index()
```

Out[91]:

	weather	avg_temp
0	Rainy	55.000000
1	Snowy	30.000000
2	Sunny	68.946271

In [96]:
```
df
```

Out[96]:

	dest	passanger	weather	temperature	time	coupon	expiration	gende
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Femal
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Femal
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Femal
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Femal
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Femal
...	...	...	...	...	...	...	...	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	Mal
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	Mal
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	Mal
12682	Work	Alone	Snowy	30	7AM	Bar	1d	Mal
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	Mal

12684 rows × 27 columns

In [104…
```
df.groupby('weather')['temperature'].count().to_frame('count').reset_index()
```

Out[104…

	weather	count
0	Rainy	1210
1	Snowy	1405
2	Sunny	10069

In [105…

```
df
```

Out[105…

	dest	passanger	weather	temperature	time	coupon	expiration	gende
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Femal
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Femal
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Femal
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Femal
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Femal
...	...	...	...	...	...	...	...	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	Mal
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	Mal
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	Mal
12682	Work	Alone	Snowy	30	7AM	Bar	1d	Mal
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	Mal

12684 rows × 27 columns

In [122…

```
df.groupby('weather')['temperature'].nunique(['temperature']).to_frame('count_disti
```

Out[122…

	weather	count_distinct_temp
0	Rainy	1
1	Snowy	1
2	Sunny	3

In [123…

```
df
```

Out[123…

	dest	passanger	weather	temperature	time	coupon	expiration	gende
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Femal
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Femal
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Femal
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Femal
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Femal
...	...	...	...	...	...	...	...	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	Mal
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	Mal
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	Mal
12682	Work	Alone	Snowy	30	7AM	Bar	1d	Mal
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	Mal

12684 rows × 27 columns

In [128…

```
df.groupby('weather')['temperature'].sum().to_frame('sum_temp').reset_index()
```

Out[128...

	weather	sum_temp
0	Rainy	66550
1	Snowy	42150
2	Sunny	694220

In [131...

```
df.groupby('weather')['temperature'].min().to_frame('min_temp').reset_index()
```

Out[131...

	weather	min_temp
0	Rainy	55
1	Snowy	30
2	Sunny	30

In [132...

```
df.groupby('weather')['temperature'].max().to_frame('min_temp').reset_index()
```

Out[132...

	weather	min_temp
0	Rainy	55
1	Snowy	30
2	Sunny	80

In [150...

```
df.groupby('occupation').filter(lambda x: x['occupation'].iloc[0] == 'Student').gro
```

Out[150...

```
occupation
Student    1584
dtype: int64
```

In [162...

```
df.groupby('occupation').filter(lambda x: x['occupation'].iloc[0]=='Student').group
```

Out[162...

```
occupation
Student    1584
dtype: int64
```

In [164...

```
df.columns
```

Out[164...

```
Index(['dest', 'passanger', 'weather', 'temperature', 'time', 'coupon',
       'expiration', 'gender', 'age', 'maritalStatus', 'has_children',
       'education', 'occupation', 'income', 'car', 'Bar', 'CoffeeHouse',
       'CarryAway', 'RestaurantLessThan20', 'Restaurant20To50',
       'toCoupon_GEQ5min', 'toCoupon_GEQ15min', 'toCoupon_GEQ25min',
       'direction_same', 'direction_opp', 'Y', 'row_count'],
      dtype='object')
```

In [190...

```
df1= pd.read_csv(r'C:\Users\pandi\Desktop\Abhiraj\Data Science\Notes\Sql Dataset\ta
```

In [167...

```
df1
```

Out[167…

	destination	passanger	weather	temperature	time	coupon	expiration	gender
0	UNION	UNION	UNION	55	2PM	Restaurant(<20)	1d	Female

1 rows × 27 columns

In [183…

```
pd.concat([df,df1])['destination'].unique()
```

Out[183…

```
array(['No Urgent Place', 'Home', 'Work', 'UNION'], dtype=object)
```

In [185…

```
pd.concat([df,df1])['destination'].drop_duplicates()
```

Out[185…

```
0     No Urgent Place
13               Home
16               Work
0               UNION
Name: destination, dtype: object
```

In [191…

```
df2= pd.read_csv(r'C:\Users\pandi\Desktop\Abhiraj\Data Science\Notes\Sql Dataset\ta
```

In [199…

```
pd.merge(df,df2[['part_of_day','time']], on= 'time',how = 'inner')[['destination','
```

Out[199…

	destination	time	part_of_day
0	No Urgent Place	2PM	Afternoon
1	No Urgent Place	10AM	Morning
2	No Urgent Place	10AM	Morning
3	No Urgent Place	2PM	Afternoon
4	No Urgent Place	2PM	Afternoon
...	...	...	...
12679	Home	6PM	Evening
12680	Work	7AM	Morning
12681	Work	7AM	Morning
12682	Work	7AM	Morning
12683	Work	7AM	Morning

12684 rows × 3 columns

In [200…

```
df.columns
```

Out[200…

```
Index(['destination', 'passanger', 'weather', 'temperature', 'time', 'coupon',
       'expiration', 'gender', 'age', 'maritalStatus', 'has_children',
       'education', 'occupation', 'income', 'car', 'Bar', 'CoffeeHouse',
       'CarryAway', 'RestaurantLessThan20', 'Restaurant20To50',
       'toCoupon_GEQ5min', 'toCoupon_GEQ15min', 'toCoupon_GEQ25min',
       'direction_same', 'direction_opp', 'Y', 'row_count'],
      dtype='object')
```

In [212…

```
df[df['passanger'] == 'Alone'][['destination' ,'passanger']]
```

Out[212…

	destination	passanger
0	No Urgent Place	Alone
1	No Urgent Place	Alone
2	No Urgent Place	Alone
3	No Urgent Place	Alone
4	No Urgent Place	Alone
...	...	...
12679	Home	Alone
12680	Work	Alone
12681	Work	Alone
12682	Work	Alone
12683	Work	Alone

12684 rows × 2 columns

In [219…

```
df[df['weather'].str.startswith('Sun')]
```

Out[219…

	destination	passanger	weather	temperature	time	coupon	expiration	g
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	F
1	No Urgent Place	Alone	Sunny	80	10AM	Coffee House	2h	F
2	No Urgent Place	Alone	Sunny	80	10AM	Carry out & Take away	2h	F
3	No Urgent Place	Alone	Sunny	80	2PM	Coffee House	2h	F
4	No Urgent Place	Alone	Sunny	80	2PM	Coffee House	1d	F
...	...	...	...	...	...	...	...	
12673	Home	Alone	Sunny	30	6PM	Carry out & Take away	1d	
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)	1d	
12677	Home	Alone	Sunny	30	6PM	Restaurant(<20)	1d	
12678	Home	Alone	Sunny	30	10PM	Restaurant(<20)	2h	
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	

10069 rows × 27 columns

In [228…

```
df[(df['temperature'] >29) & (df['temperature'] < 75)]['temperature'].unique()
```

Out[228…

```
array([55, 30])
```

In [245…

```
df[df['occupation'].isin(['Sales & Related','Management'])]['occupation']
```

Out[245…

```
193      Sales & Related
194      Sales & Related
195      Sales & Related
196      Sales & Related
197      Sales & Related
              ...
12679    Sales & Related
12680    Sales & Related
12681    Sales & Related
12682    Sales & Related
12683    Sales & Related
Name: occupation, Length: 1931, dtype: object
```

In [ ]:

In [ ]:

In [ ]:

In [ ]: